

Onychophora (110 species)



An onychophoran

Onychophorans, also called velvet worms, originated during the Cambrian explosion (see Concept 25.3). Originally, they thrived in the ocean, but at some point they succeeded in colonizing land. Today they live only in humid forests. Onychophorans have fleshy antennae and several dozen pairs of saclike legs.

Nematoda (25,000 species)



A roundworm

Also called roundworms, nematodes are enormously abundant and diverse in the soil and in aquatic habitats; many species parasitize plants and animals. Their most distinctive feature is a tough cuticle that coats their body (see Concept 33.4).

Tardigrada (800 species)

Tardigrades (from the Latin *tardus*, slow, and *gradus*, step) are sometimes called water bears for their overall shape and lumbering, bearlike gait. Most tardigrades are less than 0.5 mm in length. Some live in oceans or fresh water, while others live on plants or animals. Harsh conditions may cause tardigrades to enter a state of dormancy; while dormant, they can survive for days at temperatures as low as -200°C ! A recent phylogenomic study found that over 15% of tardigrade genes entered their genome by horizontal gene transfer, the largest fraction known for any animal.



Tardigrades (colorized SEM)

Arthropoda (1,000,000 species)

The vast majority of known animal species, including insects, crustaceans, and arachnids, are arthropods. All arthropods have a segmented exoskeleton and jointed appendages (see Concept 33.4).



A spider
(an arachnid)

Deuterostomia

Hemichordata (85 species)



An acorn worm

Like echinoderms and chordates, hemichordates are members of the deuterostome clade (see Figure 32.11). Hemichordates share some traits with chordates, such as gill slits and a dorsal nerve cord. The largest group of hemichordates is the enteropneusts,

or acorn worms. Acorn worms are marine and generally live buried in mud or under rocks; they may grow to more than 2 m in length.

Chordata (60,000 species)

More than 90% of all known chordate species have backbones (and thus are vertebrates). However, the phylum Chordata also includes two groups of invertebrates: lancelets and tunicates. See Chapter 34 for a full discussion of this phylum.



A tunicate

Echinodermata (7,000 species)



A sea urchin

Echinoderms, such as sand dollars, sea stars, and sea urchins, are marine animals in the deuterostome clade that are bilaterally symmetrical as larvae but not as adults. They move and feed by using a network of internal canals to pump water to different parts of their body (see Concept 33.5).